

longer.

According to the press release, the Philips submission will now undergo comprehensive evaluation, including performance testing conducted by independent laboratories, field assessments conducted with utilities and other partners, long-term lumen maintenance testing and stress testing under extreme conditions. Consumer pricing and retail availability are yet to be determined.

Reichl shares, "The OSRAM Parathom Classic is the very first bulb-shaped LED lamp, which constitutes a true substitute for a 40W incandescent lamp. This move will speed up the advance of LED technology as a genuine alternative for general lighting applications." The company is also aiming to develop and market an LED for SSL applications that offer a continuous spectrum, exactly like natural daylight.

The price of LED lamps is still prohibitive when one considers using them for the lighting needs of an entire house. Leading brands might cost more than \$20 a piece, while Chinese alternatives are cheaper. Prices will inevitably fall, because the writing on the wall is clear—LEDs are the future of home lighting. Despite the high initial costs, commercial organisations worldwide have already started adopting LED-based lighting because of the low TCO it offers in the long run, as well as to meet energy-related regulations.

India will also walk in step with the LED revolution happening worldwide. "We strongly believe that standardisation of LED lighting in India is a prerequisite to prepare the market for the launch of superior products. To sustain this effort, Sujana Energy is working with bodies like the Bureau of Energy Efficiency (BEE) to help define LED luminaire specifications to meet global standards in India," says Chereddi.

LED-lit towns

LEDs are a great option for street lighting, because these are economically and environmentally sustainable. By



OSRAM Opto Semiconductors' Golden DRAGON LEDs in EasyLED streetlights in Finland

moving to LEDs, a town can save lots of power, avoid a lot of carbon emission and also reduce the total cost of ownership because LEDs have a long life and are practically maintenance-free.

The first example of this kind was the Italian village of Torraca, which spent 280,000 Euros to convert all its street lamps into LED lamps, reducing their energy and maintenance costs by 70 per cent. The City of Ann Arbor in the US followed suit and replaced

a few thousand streetlights. The authorities noted an energy cost saving of around \$10,000 and an additional \$40,000 in maintenance costs last year from using LED lights instead of incandescent streetlights. The city therefore plans to replace more of its streetlights by next year.

"LED-based street lighting is on the march throughout the world. There are numerous new projects at present, especially in the Asian region. In the North American Free Trade Agreement (NAFTA) region, LEDs are being used predominantly in retrofit solutions. In Europe, the focus is on completely new systems for making towns and cities more attractive, but there are also modern street lighting systems being installed with the aim of saving energy," explains Reichl.

There are many more examples of public areas going for LEDs, parking lots in Boston and the famous Times Square Ball (a crystal time ball full of lights put up at the Times Square, Manhattan, New

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